

REFERENCE [1]

BENCHMARK DATASET

DOI: 10.24432/C53890

PPG-DaLiA: A Dataset for PPG-based Heart Rate and Activity Recognition in Daily Life

UCI Machine Learning Repository Record:

Dataset Name: PPG-DaLiA

DOI: [10.24432/C53890](https://doi.org/10.24432/C53890)

UCI ID: 495

Repository URL: <https://archive.ics.uci.edu/dataset/495/ppg+dalia>

Total Instances: ~8,300,000 samples across 15 subjects

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1. Dataset Abstract & Overview

PPG-DaLiA contains data from 15 subjects wearing physiological and motion sensors, providing a PPG dataset for motion compensation and heart rate estimation in Daily Life Activities.

PPG-DaLiA contains multi-sensor physiological and motion data collected from **15 healthy subjects** (8 males, 7 females, aged 21–55) wearing both a wrist-worn wearable device (**Empatica E4**) and a medical-grade chest sensor (**RespiBAN Professional**). Data collection spanned approximately 2.5 hours per subject under real-life conditions adhering to an 8-stage naturalistic activity protocol.

2. Hardware Configuration & Sampling Specifications

Sensor Device	Placement	Signal Modality	Sampling Rate	Unit / Resolution
Empatica E4 (Wearable)	Non-dominant	3-Axis Accelerometer (ACC)	32 Hz	g (±2g range)

Sensor Device	Placement	Signal Modality	Sampling Rate	Unit / Resolution
Wristband)	wrist	Photoplethysmography (BVP / PPG)	64 Hz	Optical absorption
		Electrodermal Activity (EDA)	4 Hz	MicroSiemens (μS)
		Skin Temperature (TEMP)	4 Hz	Degrees Celsius ($^{\circ}\text{C}$)
RespiBAN Professional (Reference Chest Unit)	Chest strap	Electrocardiogram (ECG)	700 Hz	Millivolts (mV)
		Respiration (RESP)	700 Hz	Chest circumference
		3-Axis Accelerometer	700 Hz	g ($\pm 6\text{g}$ range)
		Chest Skin Temperature / EDA	700 Hz	$^{\circ}\text{C}$ / μS

3. Ground Truth Labels & Synchronization Protocol

Field Name	Sampling Rate	Description & Purpose	Classes / Values
activity	4 Hz	Human Activity Recognition (HAR) Labels Annotated with precise timestamps during the 2.5-hour protocol.	0: Transient / Other 1: Sitting 2: Stair climbing 3: Table soccer 4: Cycling 5: Driving car 6: Lunch break 7: Walking 8: Working at desk

Field Name	Sampling Rate	Description & Purpose	Classes / Values
label	0.5 Hz	Reference Heart Rate (Ground Truth HR) Calculated every 2 seconds from 8-second R-peak intervals of chest ECG.	Continuous heart rate (beats per minute - BPM)

4. Research Context in this Capstone Project

In this study (*Đồ án Cuối kỳ AIoT - Nguyễn Bách Tùng*), the PPG-DaLiA dataset serves as the benchmark foundation:

- **Input Modality:** 3-axis wrist acceleration signals (`ACC_x`, `ACC_y`, `ACC_z`) at 32 Hz.
- **Window Segmentation:** Sliding windows of 4.0 seconds (128 samples per window) with 50% overlap (64 samples step size).
- **Target Classes:** 8 valid daily physical activities (Classes 1 through 8, excluding transient Class 0).
- **Subject-Wise Partitioning:** Subjects S1 through S11 allocated to the Training set; Subjects S12 through S15 strictly reserved for the Test holdout set (LOSO-compatible evaluation).

Tài liệu đối chiếu phục vụ Đồ án Cuối kỳ AIoT — Sinh viên: Nguyễn Bách Tùng (MSSV: 23110166)
Nguồn: UCI Machine Learning Repository | Trích dẫn: [1] A. Reiss et al., PPG-DaLiA, UCI ML Repository, 2019.